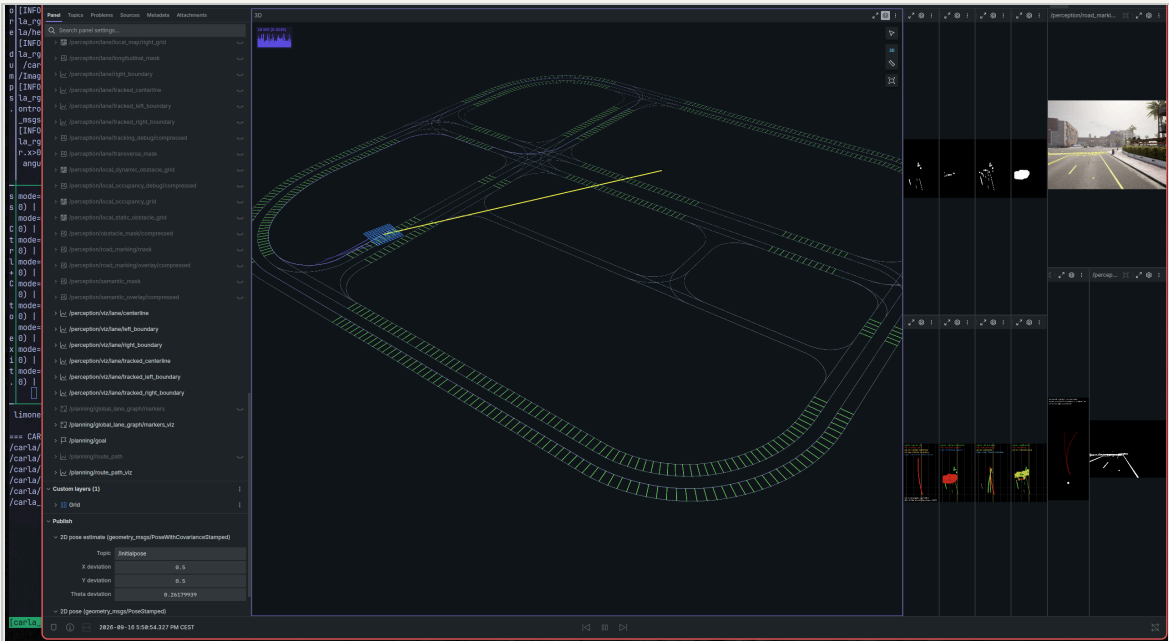


Autonomous Driving Laboratory.



GLOBAL ROUTING · FOXGLOVE

OpenDRIVE-derived routing graph, ego pose and selected RoutePlan visualized live in Foxglove.

Global routing visualization — not closed-loop autonomous driving.

Perception-derived lane mapping and custom OpenDRIVE global routing.

From perception to routing.

A modular CARLA + ROS2 prototype implementing perception-derived lane geometry, temporal lane tracking, a rolling LaneMap and custom OpenDRIVE global routing.

Individual engineering project · ROS2 / CARLA research prototype

LOCAL / LaneMap

Perception → geometry → tracking → rolling lane memory

GLOBAL / RoutePlan

OpenDRIVE → directed graph → A^* → nominal global route

NEXT — NOT IMPLEMENTED

RoutePlan ↔ LaneMap association

IMPLEMENTED

Lane perception

IMPLEMENTED

Temporal LaneMap

IMPLEMENTED

Global routing

NEXT

Route/lane association

FUTURE

Trajectory planning

FUTURE

Tracking / control

Daniel Martínez-Cabeza de Vaca Guillén

Dual degree in Mathematics and Computer Science · University of Murcia

Source: github.com/DonFisto/vision-segmentation-autonomous-driving

dmcvg@icloud.com

github.com/DonFisto

Global topology and local evidence stay separate.

01 / SENSOR-DERIVED GEOMETRY

Local perception

IMPLEMENTED

- ↓ CARLA RGB + CameraInfo
- ↓ Binary road-marking segmentation
- ↓ Metric BEV lane geometry
- ↓ Geometric / component / context filters
- ↓ Quadratic curve fitting
- ↓ Temporal tracking + hero odometry

LaneMap

Rolling, local perception memory

02 / PRIVILEGED SIMULATOR MAP

Global map / routing

IMPLEMENTED

- ↓ CARLA / OpenDRIVE
- ↓ Directed road / lane topology
- ↓ Sampled directed routing graph
- ↓ Legal lane-follow / lane-change edges
- ↓ Ego + goal graph association
- ↓ A* search · Dijkstra baseline

RoutePlan

Global, nominal route geometry

NEXT · NOT IMPLEMENTED

RoutePlan ↔ LaneMap association

Global intent meets local perception — through geometry, not ID equality.

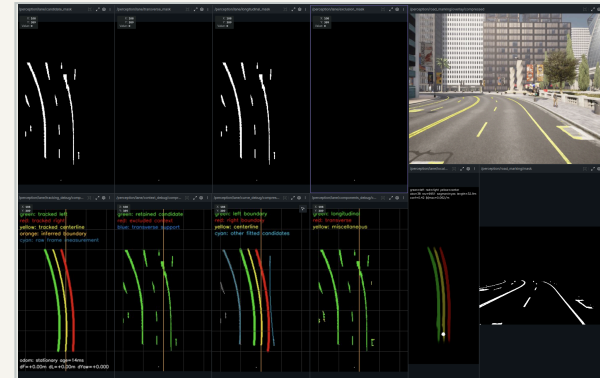
FUTURE Behavior / maneuver planning

FUTURE Local trajectory planning

FUTURE Trajectory tracking / control

— Solid: implemented within each stream — Dashed: planned integration

OpenDRIVE is privileged map information, not perception evidence.



LANE PERCEPTION · TEMPORAL TRACKING

Road-marking candidates, filtered curves and temporal lane tracking alongside camera context.

This pipeline feeds the rolling LaneMap; the diagnostic view does not display every LaneMap field.

Implemented lane chain

1. Detect road markings
2. Recover metric geometry
3. Filter geometric components
4. Reject misleading context
5. Fit lane curves
6. Track through ego motion

Rolling vector LaneMap

Sampled boundaries and centerline, widths, curvature and confidence. Schema 2 emits zero or one local segment; positions/headings publish in direct `carla.world`.

Only accepted direct observations become fresh memory. Held/inferred tracks do not reinforce themselves. Topology and speed limit stay unknown.

With fresh odometry but insufficient perception, an empty LaneMap is valid. This is local memory, not OpenDRIVE or a globally consistent map.

Source record: lane perception, tracking & mapping ↗

A global route is nominal intent, not a trajectory.

- 01

Extract directed topology
CARLA/OpenDRIVE supplies coarse road/lane connectivity and sampled edge geometry.
- 02

Build the explicit graph
Longitudinal samples form lane-follow edges. Permitted left/right transitions connect adjacent sampled driving lanes outside junctions.
- 03

Associate ego and goal
World positions project onto a driving lane, then a sampled graph node, with distance and longitudinal-error rejection gates.
- 04

Search with a reference
Dijkstra provides the baseline/oracle. Production A* uses Euclidean distance with nonnegative edge costs and lane-change penalties.
- 05

Publish RoutePlan
Successful search publishes VALID. A completed no-path search publishes INVALID with empty geometry. This is nominal global routing, without trajectory dynamics.

Observed in development validation

Town10HD / Town10HD_Opt, CARLA 0.9.16: 3,066 sampled nodes, 4,522 edges and one strongly connected component.

One frame probe: mean lane-to-route distance 2.949 m direct versus 51.123 m with y flipped. Supports coordinate convention, not association accuracy.

Reported September 2026; not rerun for this portfolio. Raw probes/bags are not checked in. Scenario observations, not performance guarantees.

Production interfaces · current lane and routing path

TOPIC / MESSAGE TYPE	FRAME
/carla/hero_odom nav_msgs/msg/Odometry	carla_world → hero
/perception/lane/tracked_centerline nav_msgs/msg/Path	hero
/perception/lane/local_map/vector autonomy_interfaces/msg/LaneMap	carla_world
/planning/goal geometry_msgs/msg/PoseStamped	carla_world
/planning/route_plan autonomy_interfaces/msg/RoutePlan	carla_world

RoutePlan contract

Coarse graph geometry with global topology provenance. A* is production search; Dijkstra remains the reference. Completed no-path search publishes INVALID. Rejected goals retain the previous accepted goal.

Reliable, transient-local, depth-1 publication retains the latest message while the publisher lives. No route-age timeout; confidence is heuristic, not probability.

Production / display isolation

carla_world → hero production
carla_world_viz → hero_viz display only

Foxglove reflects copied global geometry. No physical TF connects the two world bases. _viz topics are not planner inputs.

[Source record: global route planning ↗](#)

Connecting global intent to local perception.

Earlier · phases 0–7

From model experiments to spatial perception

01–04 Aug 2026 · phase 8

Lane-specific perception & geometry

04–18 Aug 2026 · phase 9

Odometry-aware tracking → rolling LaneMap

20–23 Aug 2026 · phase 10

OpenDRIVE topology → graph search → RoutePlan

15–16 Sep 2026 · phase 11

Frame contracts, visualization & reproducible startup

Next · phase 12

RoutePlan ↔ LaneMap association

Engineering decisions

- Keep privileged map topology separate from perceived evidence.
- Own the routing graph; retain Dijkstra as an A* reference.
- Correct production geometry at its source; reflect only display copies.
- Keep global/local IDs and revision meanings independent.
- Record bounded runtime evidence before closing integration milestones.

Daniel Martínez-Cabeza de Vaca Guillén

Dual degree in Mathematics and Computer Science · University of Murcia
Expected graduation June 2028 · Erasmus, Universität Stuttgart 2026/2027

Spanish — Native · English — C1 · German — B1 · French — Basic

Reviewed 17 September 2026 · [feature/route-lane-association](#) · [42a250e34197](#) · [donfisto.github.io](#)

NEXT · NOT IMPLEMENTED

RoutePlan ↔ LaneMap association

Planned: ego-local route window, projection, progress s , signed offset d , heading discrepancy, overlap/coverage, confidence gates and explicit no-valid-association output. Geometry, not ID equality.

Current boundaries

- Association, behavior/maneuver planning, local trajectory planning and trajectory tracking/control are not implemented for this pipeline.
- CARLA odometry and OpenDRIVE remain privileged inputs. The graph is static, without dynamic-obstacle/prediction costs. Goal orientation is ignored.
- Perception may yield an empty LaneMap. Downstream signed-curvature semantics remain unvalidated.

FUTURE · Simple local reference/trajectory planner → understandable baseline tracking/control → repeatable closed-loop evaluation.

Open to working-student roles and robotics or autonomous-driving opportunities.

dmcvg@icloud.com · github.com/DonFisto